

Biology

1. The substances that connect light reaction and dark reaction are ATP and NADPH as they are products of light reaction which are then used for dark reaction.
2. In aerobic respiration eukaryotic cells aerobic respiration takes place in mitochondria.
3. The G lymph is devoid of A.B.C. (red blood cells).
4. Plants get rid of excess water by the process of transpiration.
5. The malpighian body is an important part of nephron formed of glomerulus and Bowman's capsule.
6. The following factors do not contribute to rate of transpiration:
 - (a) Oxygen level
 - (b) Chlorophyll content
 - (c) Age of plant

Respiration in Plants	Respiration in Animals
① Plants do not breathe, they only perform respiration.	① Animals have lungs for further cellular respiration.
② They don't have respiratory system.	② Lysate respiratory system is found.
③ They take in O_2 through root, stomata, cuticle.	③ It occurs through lungs.
④ Rate of respiration is slow.	④ Rate of respiration is fast.

8 Light Reaction

- ① It takes place in thylakoids
- ② Depends on sunlight
- ③ Oxygen is liberated
- ④ ATP and NADPH are produced
- ⑤ Requires chlorophyll

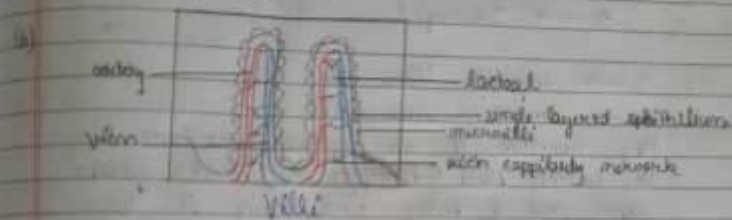
Dark Reaction

- ① It takes place in stroma
- ② Does not depend on sunlight
- ③ CO_2 is fixed
- ④ Glucose is produced using energy from ATP and NADPH
- ⑤ Does not require chlorophyll

9 (a) Enzymes which act on proteins are pepsin and trypsin.

(b) The fate of pyruvate depends on the availability of oxygen. In the presence of O_2 , pyruvate is transported to mitochondria for further energy production but in absence of O_2 , pyruvate is converted to ethyl alcohol or lactic acid.

- Q (a) The role of hydrochloric acid (HCl) in stomach is to make the environment acidic which helps in activation of certain digestive enzymes like pepsinogen, trypsinogen, chymotrypsinogen and also helps in killing certain harmful microorganisms.



Lacteal present in villi are similar like the structures which function where fat is absorbed. Fats from small intestine are absorbed as fatty acids & glycerol.

- Q (a) Alveoli present in the lungs are covered with blood capillaries so that the exchange of ~~from~~ air/gases can take place among alveoli membrane and blood capillaries. The blood present in blood have poor concentration of O_2 while rich concentration of CO_2 so on the other hand the membrane of alveoli have rich concentration of O_2 and low concentration of CO_2 . The difference in concentration allows O_2 to diffuse from air capillaries and CO_2 to diffuse out of blood.

(n) The walls of trachea are surrounded by several rings cartilaginous ring to provide flexibility to trachea. This also enables trachea not to collapse while there is air / no air in trachea.

13. (i) 4 → Mitral Valve (Bicuspid)
5 → Tricuspid

(ii) 3 → Superior Vena Cava

6 → Inferior Vena Cava

The function of ~~the~~ superior Vena Cava is to collect blood (deoxygenated blood) from upper part of body and bring it to a right atrium while the function of inferior Vena Cava is to bring blood from lower part of body to right atrium.

(iii) The blood vessels that carry away blood from heart are arteries.

(iv) Circulation in humans is double circulation because the ~~the~~ blood flows / passes through the heart twice which is systemic circulation and other is pulmonary circulation.

Q2 (a) In winter

(b) Accessory excretory organs in humans are lungs, skin, liver and large intestine

(c) Different types of excretory organs found in arthropods are:

- ① Insects like cockroach $\rightarrow$ Malpighian tubules
- ② Earthworm $\rightarrow$ Nephridia

(d) In winter we ~~excrete~~ less urine to lead for metabolism ~~also~~ due to the low concentration of ADH in blood and less permeability of DCT which results in less reabsorption of H_2O in capillaries which after all results in more urine formation of urine